

TECNOLOGÍAS PARA LA DIGITALIZACIÓN

Mathematical Engineering and Artificial Intelligence

Lab Practice 2: Identity and Access Management and cost estimation in Google Cloud

Lab Report

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1 Objective

This lab covers Identity and Access Management in Google Cloud: predefined roles, custom roles, service accounts, roles granted at resource level instead of project level, and a least-privilege design for a realistic scenario. It ends with a cost estimation of the deployed setup using the Google Cloud Pricing Calculator.

Our team is made of two students and each of us has used his own Google account. User-A is Andrés Gil, who owns the project lab-2-a (ID lab-2-a-508905), and User-B is Jorge Carnicero (gfxpollo@gmail.com), who works on that same project with the roles User-A grants him along the lab.

The screenshots that we show are taken from whichever account performed each step and the caption says which one when it is not obvious. In terms of regions, everything was created in europe-west1.

2 Task 1 – Predefined IAM roles

2.1 Project creation and initial access

User-A created the project lab-2-a. Before granting anything, the IAM page showed only User-A, with the roles inherited from his organization.

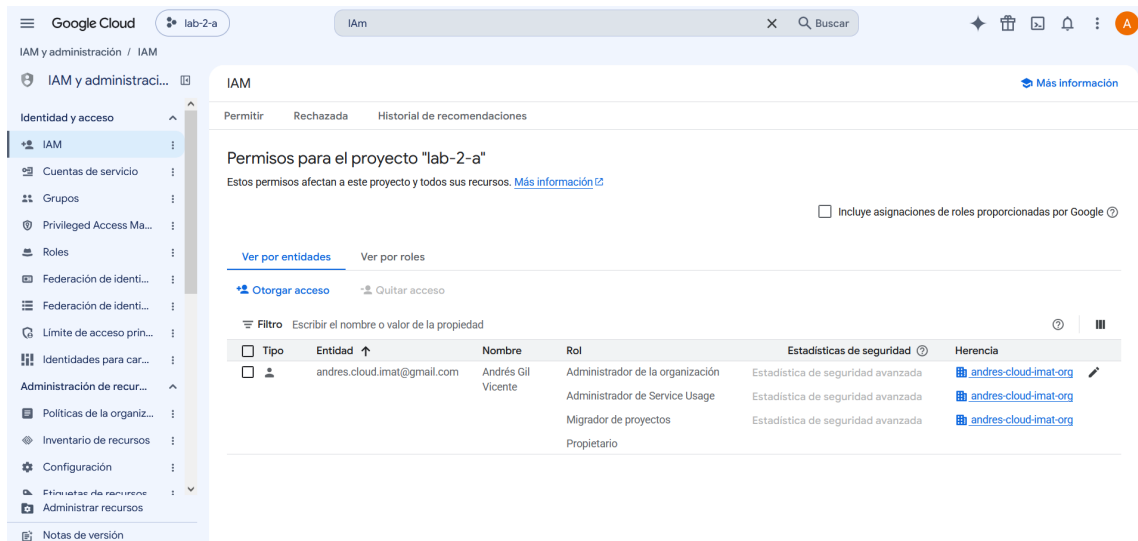


Figure 1: IAM page of lab-2-a before granting access to User-B

Q: Which are the required predefined IAM roles for these tasks?

Compute Admin (roles/compute.admin), granted to User-B at project level. It gives full control of all Compute Engine resources, which is what "administer any Compute Engine instance" asks for.

Otorgar acceso a "lab-2-a"

Grant principals access to this resource and add roles to specify what actions the principals can take. Optionally, add conditions to grant access to principals only when a specific criteria is met. [Más información sobre las condiciones de IAM](#)

Recurso

lab-2-a

Agregar entidades

Las principales son usuarios, grupos, dominios o cuentas de servicio. [Más información sobre las principales de IAM](#)

Entidades nuevas *

gfxpollo@gmail.com X ?

Asignar roles

Los roles se componen de conjuntos de permisos y determinan lo que la principal puede hacer con este recurso. [Más información](#)

Rol *

Administrador de Compute ▼

Control absoluto de todos los recursos de Compute Engine.

Condición de IAM (opcional) ?

+ Agregar condición de IAM

+ Agregar otro rol

Guardar **Cancelar**

Figure 2: User-A grants *Compute Admin* to User-B on lab-2-a

After the grant, User-B could already see the project in the resource selector, choosing *No organization* and then *All projects*, since the project belongs to User-A's organization.

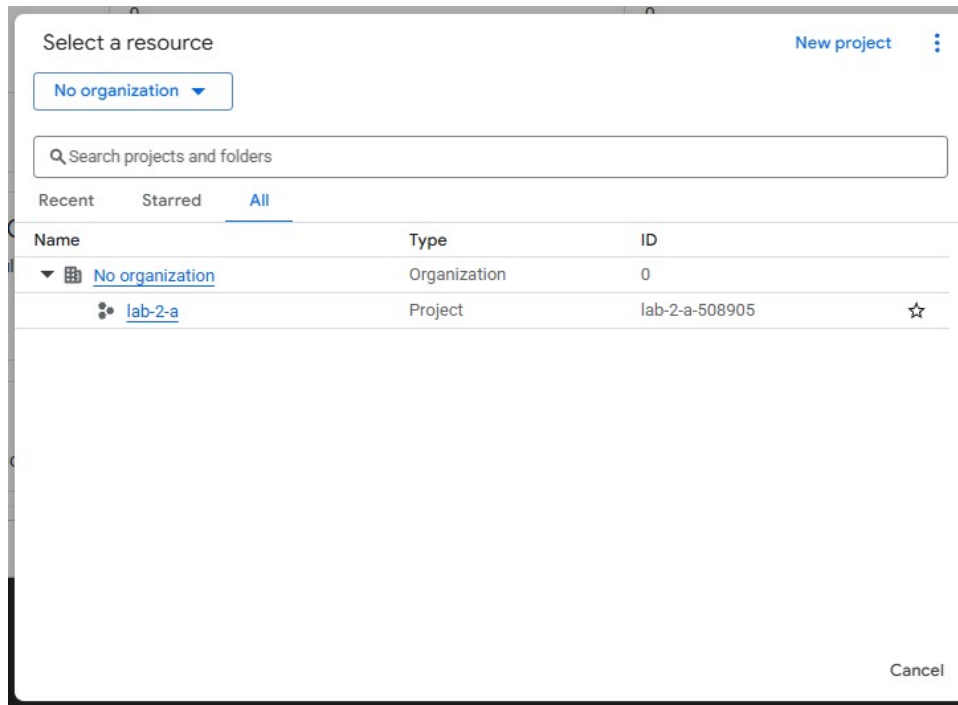


Figure 3: User-B selects lab-2-a from the resource selector

2.2 VM2 and the SSH firewall rule

The project was new, so the `ml-network` VPC of Lab 1 did not exist here and had to be created again with the same configuration (custom mode, subnet `ml-subnetwork` `10.0.0.0/24` in `eu-west-1`). Then User-A created the instance.

Name	vm2
Machine type	e2-medium
Region / zone	eu-west-1 / eu-west-1-c
Image	Debian GNU/Linux 13 (trixie), the console default
Boot disk	Balanced persistent disk, 10 GB
Network tag	deep
Network	ml-network / ml-subnetwork

Table 1: VM2 configuration

The statement asks to leave the default Debian 12, but the default image offered by the console was already Debian 13 (trixie), so that is what we used.

Boot disk

Select an image or snapshot to create a boot disk; or attach an existing disk. Can't find what you're looking for? Explore hundreds of VM solutions in [Marketplace](#)

[Public images](#) Custom images Snapshots Archive Snapshots Existing Disks

Operating system
Debian

Version *
Debian GNU/Linux 13 (trixie)
x86/64, amd64 built on 20260908

Boot disk type *
Balanced persistent disk

[Compare disk types](#)

Size (GB) *
10
Provision between 10 and 65536 GB

[Show advanced configuration](#)

[Select](#) [Cancel](#)

(a) Default boot disk image

← Create an instance [Create VM from...](#)

nic0 default default IPv4 (10.132.0.0/20)

Machine configuration
e2-medium, europe-west1-c

OS and storage
Debian GNU/Linux 13 (trixie)

Data protection
Snapshot schedules

Networking
2 network interfaces

Observability
Install Ops Agent

Security

Advanced

New network interface

Interface type
☒ VPC [?](#)
☐ Private Service Connect [?](#)

Network *
ml-network [?](#)

Subnetwork *
ml-subnetwork IPv4 (10.0.0.0/24) [?](#)

To use IPv6, you need an IPv6 subnet range.
[Learn more](#)

Network interface card
— [?](#)

IP stack type
☒ IPv4 (single-stack)

(b) Machine configuration and network interface

Figure 4: Creation of vm2

SSH access was allowed with the same rule as in Lab 1: ingress, TCP 22, source 35.235.240.0/20 (the IAP range), applied to all instances of the network.

lab-2-a vpc

← Create a firewall rule

Deny

Targets
All instances in the network

Source filter
IPv4 ranges

Source IPv4 ranges *
35.235.240.0/20 × for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter
None

Destination filter
None

Protocols and ports ?

☐ Allow all

☒ Specified protocols and ports

☒ TCP

Ports
22
E.g. 20, 50-60

☐ UDP

(a) Rule definition

Google Cloud lab-2-a Search (/) for resources, docs, products, and more

← Firewall rule details Edit Delete

IP ranges 35.235.240.0/20

Protocols and ports
tcp:22

Enforcement
Enabled

Insights
None

Hit count monitoring ?
—

Applicable to instances

i The following table does not show any App Engine flexible environment instances

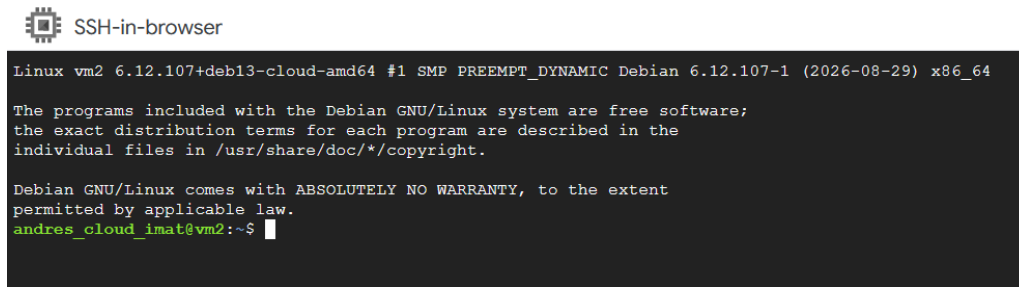
Filter Filter by instance name, project or subnetwork

Name ↑	Subnetwork	Internal IP ranges	External IP ranges	Tags	Service accounts
vm2	default	10.132.0.2	34.140.164.242	deep	737974638829-

(b) The rule applies to vm2

Figure 5: SSH firewall rule from the IAP range

With the rule in place, User-A started the instance and connected from the browser without any problem.



SSH-in-browser

```
Linux vm2 6.12.107+deb13-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

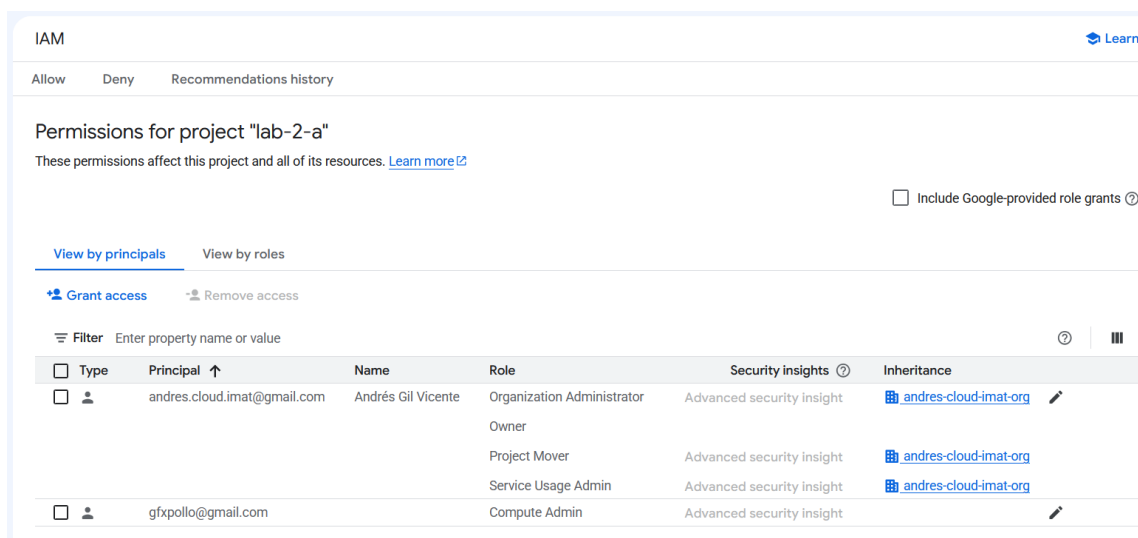
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
andres_cloud_imat@vm2:~$
```

Figure 6: User-A connected to vm2 through SSH-in-browser

Q: What is the IAM role that is assigned to User-A? What type of role is it (basic/predefined/custom)?

User-A holds **Owner** (roles/owner) on lab-2-a, which is a **basic** role: the three basic roles are Owner, Editor and Viewer, they existed before IAM and they apply to almost every service in the project.

The IAM page also lists three predefined roles that User-A does not hold on the project itself but inherits from the organization **andres-cloud-imat-org**: Organization Administrator, Project Mover and Service Usage Admin.



IAM

Allow Deny Recommendations history

Permissions for project "lab-2-a"

These permissions affect this project and all of its resources. [Learn more](#)

☐ Include Google-provided role grants

[View by principals](#) [View by roles](#)

[Grant access](#) [Remove access](#)

Filter Enter property name or value

Type	Principal	Name	Role	Security insights	Inheritance
☐	andres.cloud.imat@gmail.com	Andrés Gil Vicente	Organization Administrator	Advanced security insight	andres-cloud-imat-org
			Owner		
			Project Mover	Advanced security insight	andres-cloud-imat-org
			Service Usage Admin	Advanced security insight	andres-cloud-imat-org
☐	gfxpollo@gmail.com		Compute Admin	Advanced security insight	

Figure 7: Roles of User-A (Owner, plus roles inherited from the organization) and of User-B (Compute Admin)

2.3 Cloud Storage bucket

User-A created the bucket `imat-lab2-bucket` and uploads a file (`TD-lab2.pdf`) to it. The bucket is a regional bucket with storage class `Standard` and not public.

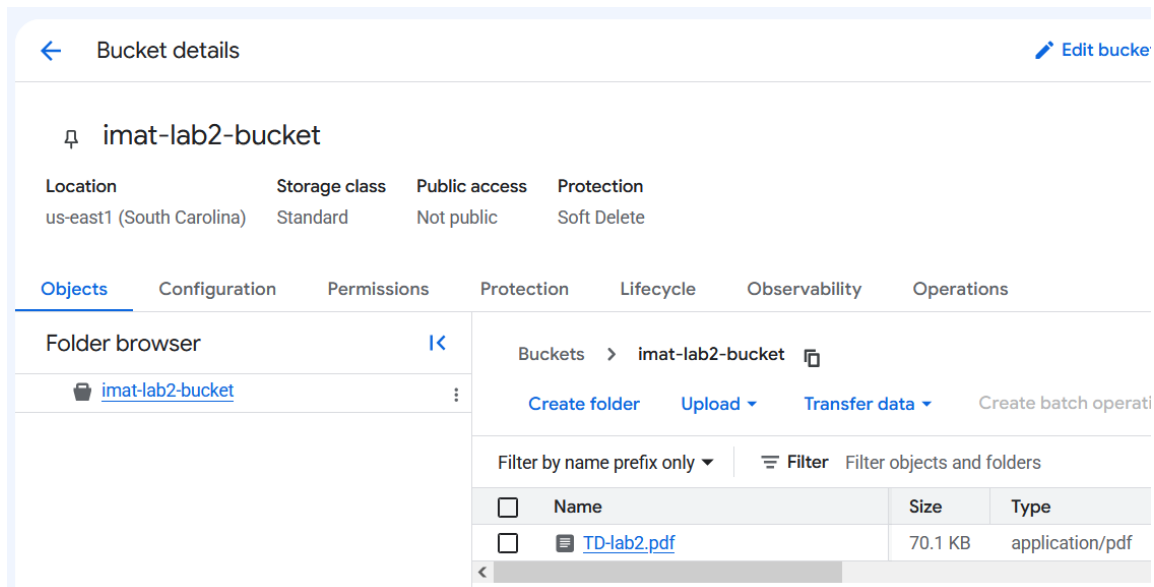


Figure 8: Bucket `imat-lab2-bucket` with the uploaded file

2.4 User-B administers the VM

Q: User-B inspects the vm configuration and changes the disk size, then connects to it. Are these actions allowed? Why? Why not?

This is partly allowed. Inspecting the instance and resizing its boot disk from 10 GB to 11 GB worked, because `roles/compute.admin` contains the permissions needed for that (`compute.instances.get`, `compute.disks.update` and so on).

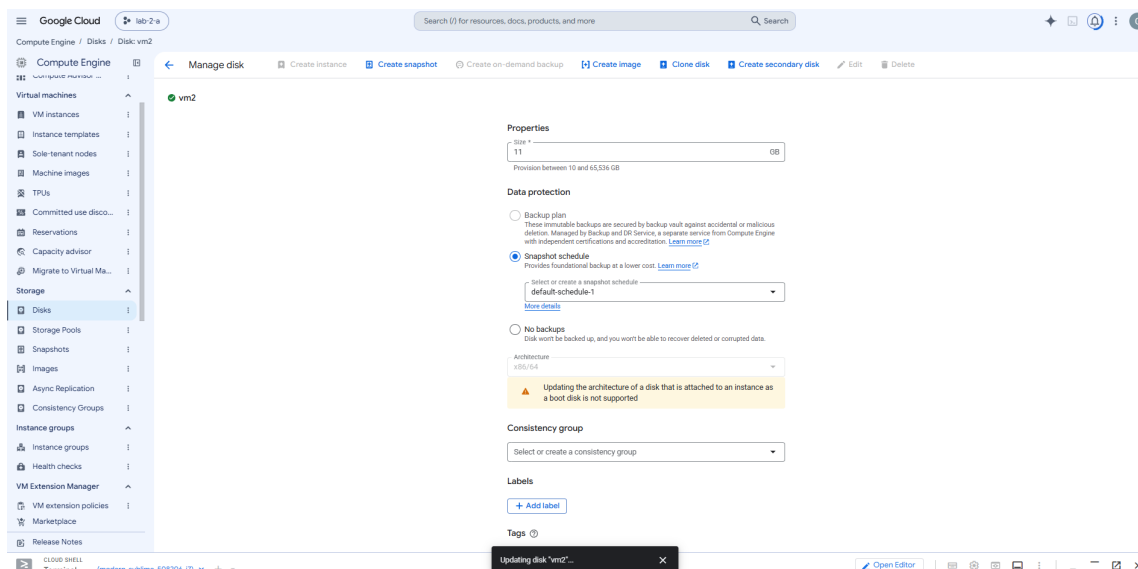


Figure 9: User-B changes the disk size of `vm2` from 10 to 11 GB

Connecting, however, was rejected. The console reported that User-B lacks:

```
roles/iam.serviceAccountUser
```

The reason is that `vm2` runs as the default Compute Engine service account, and SSH-ing into it means using that identity, so IAM requires the Service Account User role on top of Compute Admin.

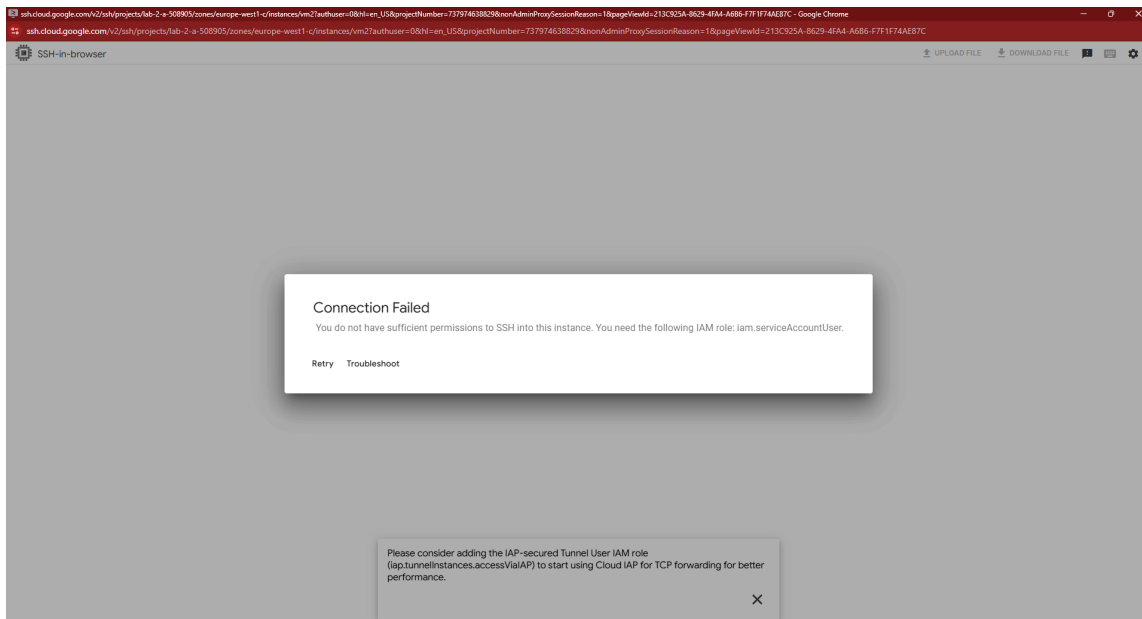


Figure 10: SSH denied: the missing role is `iam.serviceAccountUser`

2.5 User-B lists the buckets

Q: User-B lists the Cloud Storage buckets. Is this action allowed? Why? Why not?

No, User-B only has the Compute Admin role, which contains no Cloud Storage permissions. The console showed the missing permission explicitly, `storage.buckets.list`, and suggested the roles that would provide it.

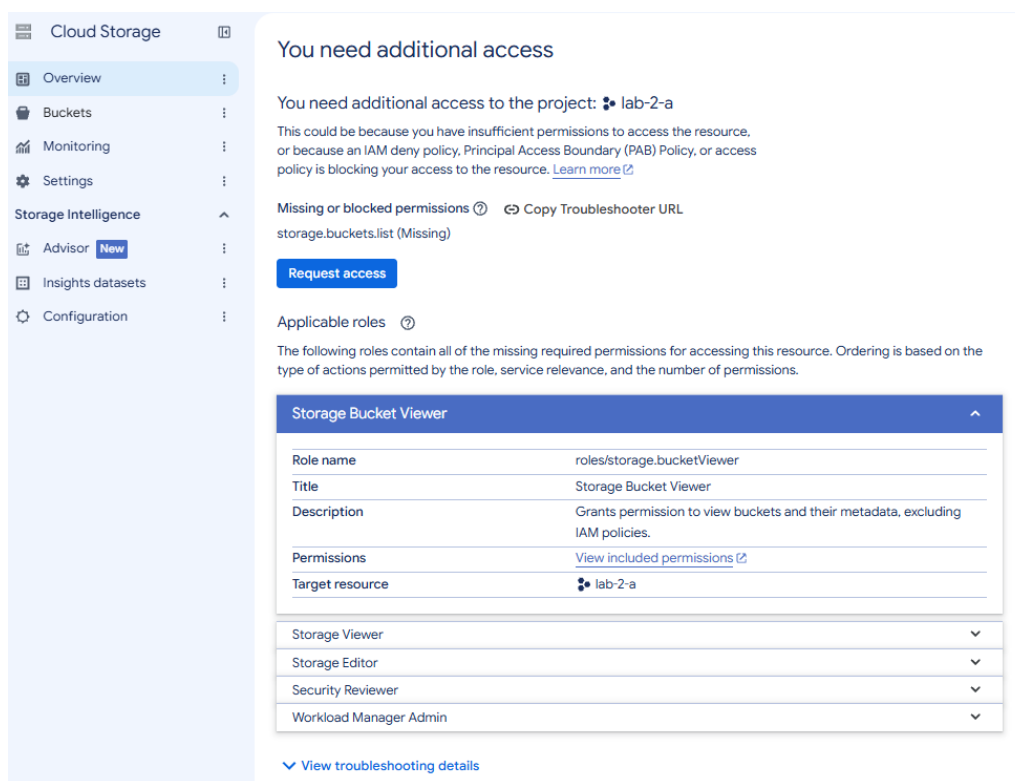


Figure 11: User-B cannot list the buckets: `storage.buckets.list` is missing

2.6 Granting access to the buckets

Q: Which are the required predefined IAM roles for these tasks?

Storage Bucket Viewer (`roles/storage.bucketViewer`), which grants the permissions `storage.buckets.get` and `storage.buckets.list`, nothing else. It is the smallest predefined role that solves the problem, so it is the right one under least privilege; `roles/storage.admin` or `roles/viewer` would also have worked but would have granted permissions for far more than listing buckets.

Storage Bucket Viewer Beta 
(`roles/storage.bucketViewer`)

`storage.buckets.get`
`storage.buckets.list`

Grants permission to view buckets and their metadata,
excluding IAM policies.

Figure 12: Storage Bucket Viewer Role

Grant access to "lab-2-a"

Grant principals access to this resource and add roles to specify what actions the principals can take. Optionally, add conditions to grant access to principals only when a specific criteria is met. [Learn more about IAM conditions](#)

Resource

lab-2-a

Add principals

Principals are users, groups, domains, or service accounts. [Learn more about principals in IAM](#)

New principals *

gfxpollo@gmail.com ✕ ?

Assign roles

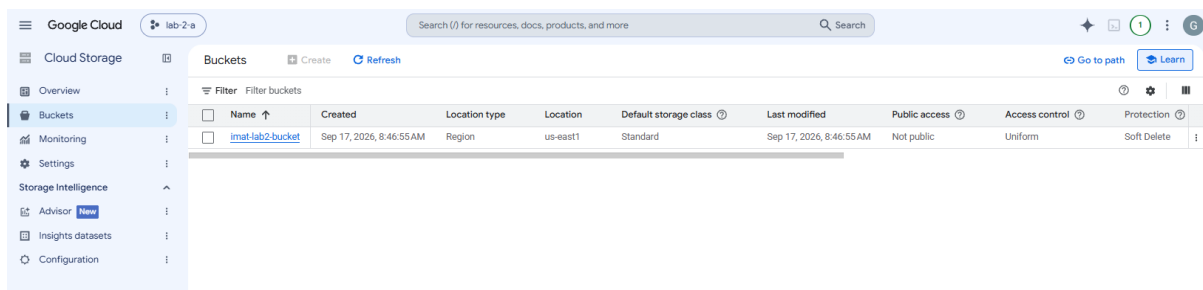
Roles are composed of sets of permissions and determine what the principal can do with this resource. [Learn more](#)

Role * IAM condition (optional) ?

Storage Bucket Viewer (Beta) ▼ + Add IAM condition

Grants permission to view buckets and their metadata, excluding IAM policies.

(a) User-A grants the role



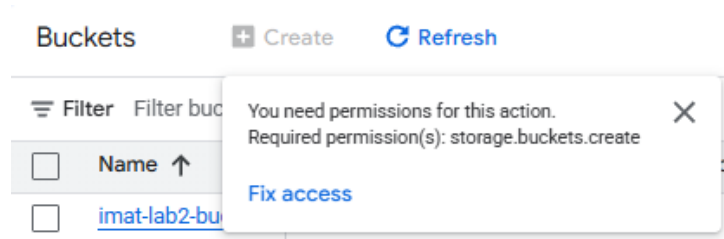
(b) User-B can now list the bucket

Figure 13: Storage Bucket Viewer granted to User-B and the result

2.7 Create new bucket & User-B tries to grant himself the permission

Q: User-B want to create a bucket but he can't so User-B tries to assign the permission on his own. What is the result?

It fails. Creating a bucket needs `storage.buckets.create`, which User-B does not have, and when he tries to fix it himself from the IAM page the console answers that `resourcemanager.projects.getIamPolicy` is missing. Modifying an allow policy requires permissions on the policy itself (`getIamPolicy` and `setIamPolicy`), and having a role does not let anyone hand out roles. This is what prevents a user from escalating his own privileges by default.



(a) Creating a bucket needs `storage.buckets.create`

You need additional access

You need additional access to the project: lab-2-a

This could be because you have insufficient permissions to access the resource, or because an IAM deny policy, Principal Access Boundary (PAB) Policy, or access policy is blocking your access to the resource. [Learn more](#)

Missing or blocked permissions Copy Troubleshooter URL
resourcemanager.projects.getIamPolicy (Missing)

[Request access](#)

Applicable roles

The following roles contain all of the missing required permissions for accessing this resource. Ordering is based on the type of actions permitted by the role, service relevance, and the number of permissions.

Role Viewer	
Role name	roles/iam.roleViewer
Title	Role Viewer
Description	Read access to all custom roles in the project.
Permissions	View included permissions
Target resource	lab-2-a
Security Reviewer	
Data Studio Asset Editor	
Workload Manager Admin	
Firebase Viewer	

[View troubleshooting details](#)

(b) And editing the policy needs `resourcemanager.projects.getIamPolicy`

Figure 14: User-B cannot grant himself the missing permission

2.8 Granting User-B the right to manage the policy

Q: Which are the required predefined IAM roles for these tasks?

IAM Admin (roles/iam.admin) and Project IAM Admin (roles/resourcemanager.projectIamAdmin). The first one manages custom roles and service accounts in the project. The second one allows getting and setting the allow policy of the project, that is, granting roles to principals.

Grant access to "lab-2-a"

Grant principals access to this resource and add roles to specify what actions the principals can take. Optionally, add conditions to grant access to principals only when a specific criteria is met. [Learn more about IAM conditions](#)

Resource

lab-2-a

Add principals

Principals are users, groups, domains, or service accounts. [Learn more about principals in IAM](#)

New principals *

gfxpollo@gmail.com X ?

Assign roles

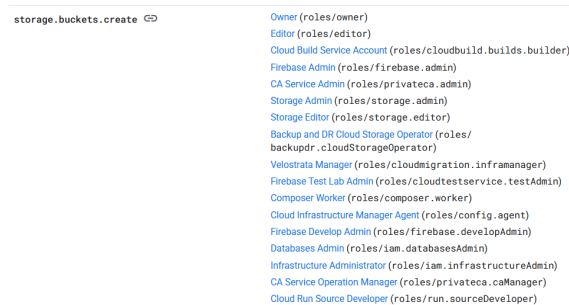
Roles are composed of sets of permissions and determine what the principal can do with this resource. [Learn more](#)

Role *	IAM condition (optional) ?	
Iam Admin	+ Add IAM condition	
Admin role for iam		
Role	IAM condition (optional) ?	
Project IAM Admin	+ Add IAM condition	
Access and administer a project IAM policies.		

Figure 15: User-A grants IAM Admin and Project IAM Admin to User-B

2.9 User-B grants himself the bucket permissions

With those roles, User-B looked up which roles contain `storage.buckets.create` and picked Storage Editor (`roles/storage.editor`) rather than Storage Admin, again to avoid giving himself more than needed.



(a) Roles that contain `storage.buckets.create`

Grant access to "lab-2-a"

Grant principals access to this resource and add roles to specify what actions the principals can take. Optionally, add conditions to grant access to principals only when a specific criteria is met. [Learn more about IAM conditions](#)

Resource

lab-2-a

Add principals

Principals are users, groups, domains, or service accounts. [Learn more about principals in IAM](#)

New principals *

gfxpollo@gmail.com

Assign roles

Roles are composed of sets of permissions and determine what the principal can do with this resource. [Learn more](#)

Role *
Storage Editor
Editor role for storage

IAM condition (optional)
+ Add IAM condition

+ Add another role

Help me choose roles

Save Cancel

(b) User-B assigns himself Storage Editor

Figure 16: Self-granting the permission to create buckets

← Create a bucket

- Get started**
 Pick a globally unique, permanent name. [Naming guidelines](#)
 Ex: 'example', 'example_bucket-1', or 'example.com'
 Tip: Don't include any sensitive information
 Labels (optional) ▼
 Continue
- Choose where to store your data**
 Location: us (multiple regions in United States)
 Location type: Multi-region
- Choose how to store your data**
 Default storage class: Standard
 Hierarchical namespace: Disabled
 Rapid Cache: Disabled
- Choose how to control access to objects**
 Public access prevention: On
 Access control: Uniform
 IP filtering: Not configured
- Choose how to protect object data**
 Soft delete policy: Default
 Object versioning: Disabled
 Bucket retention policy: Disabled
 Object retention: Disabled
 Encryption type: Google-managed
 Allowed encryption rules: Google-managed and Cloud KMS

Create [Equivalent code](#) Cancel

Good to know

Location pricing
 Storage rates vary depending on the storage class of your data and location of your bucket. [Pricing details](#)

Current configuration: Multi-region / Standard

Item	Cost
us (multiple regions in United States)	\$0.026 per GB-month
With default replication	\$0.020 per GB written

[Estimate your monthly cost](#)

Figure 17: With Storage Editor, User-B can now create a bucket

Q: What do you think about this method to assign the required permissions? Is it secure?

No, it is not secure. If a user can edit the allow policy of the project, he can grant himself any role, so the effective privilege of that user is not the role he holds but everything he can give himself, up to Owner. Permissions should be granted by an authorized administrator, only for the task at hand, and preferably to groups rather than individuals.

3 Task 2 – Custom role

The requirement is to create, delete and update buckets and objects, without being able to grant permissions on them. `roles/storage.admin` covers the first part but also contains the `setIamPolicy` permissions, so we built a custom role from it.

Using *Create custom role from selection* with Storage Admin as a base, the 112 permissions of the role were preloaded and the three `setIamPolicy` permissions were deleted:

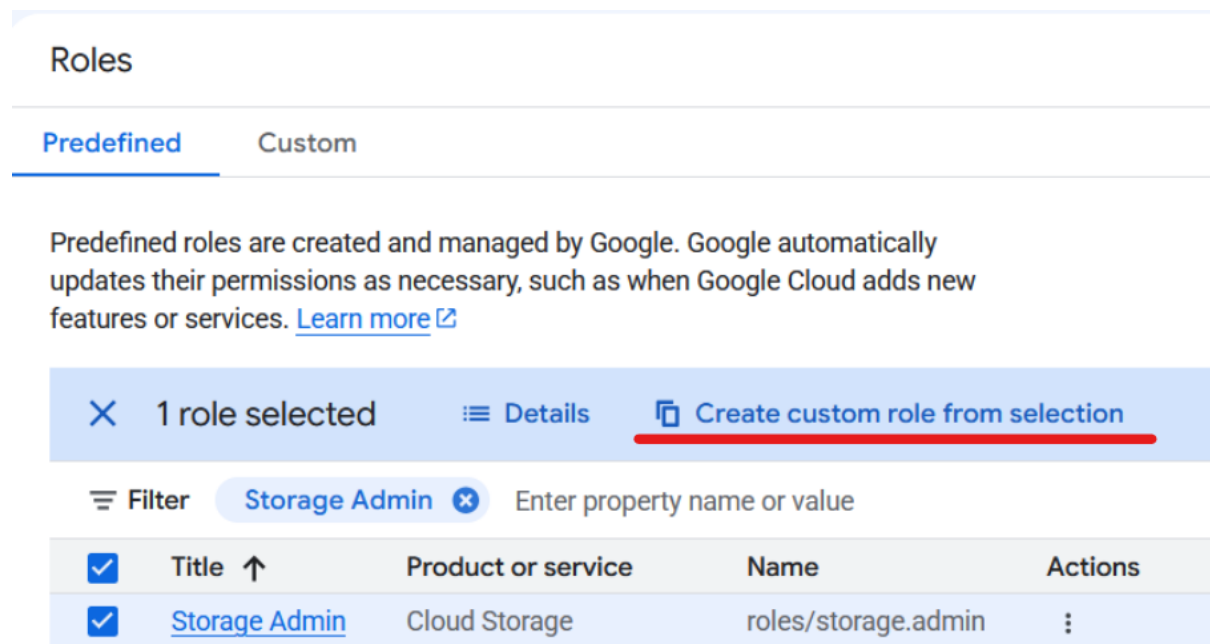


Figure 18: Create custom role from selection

Listing 3.1: Permissions removed from the custom role

```
storage.buckets.setIamPolicy
storage.managedFolders.setIamPolicy
storage.objects.setIamPolicy
```

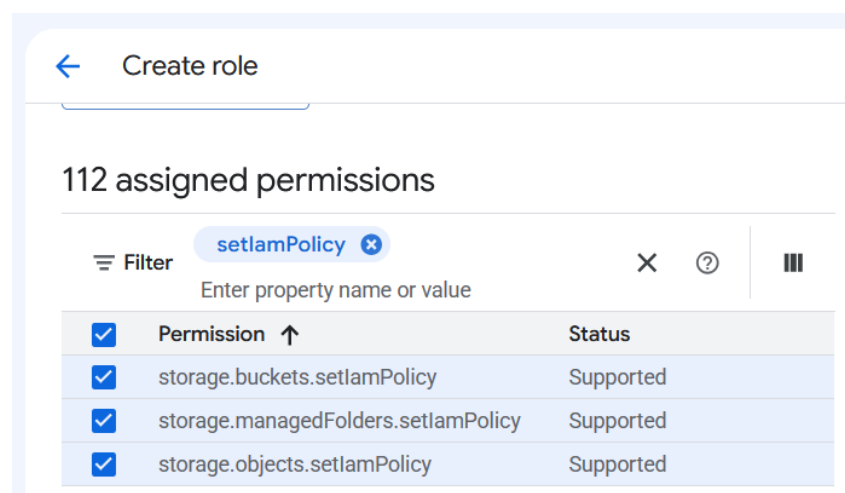


Figure 19: Filtering by setIamPolicy to unselect those permissions

The resulting role is **Storage Admin** without **IAM policies permissions**. User-A then removed **Storage Editor** from User-B and assigned the custom role instead:

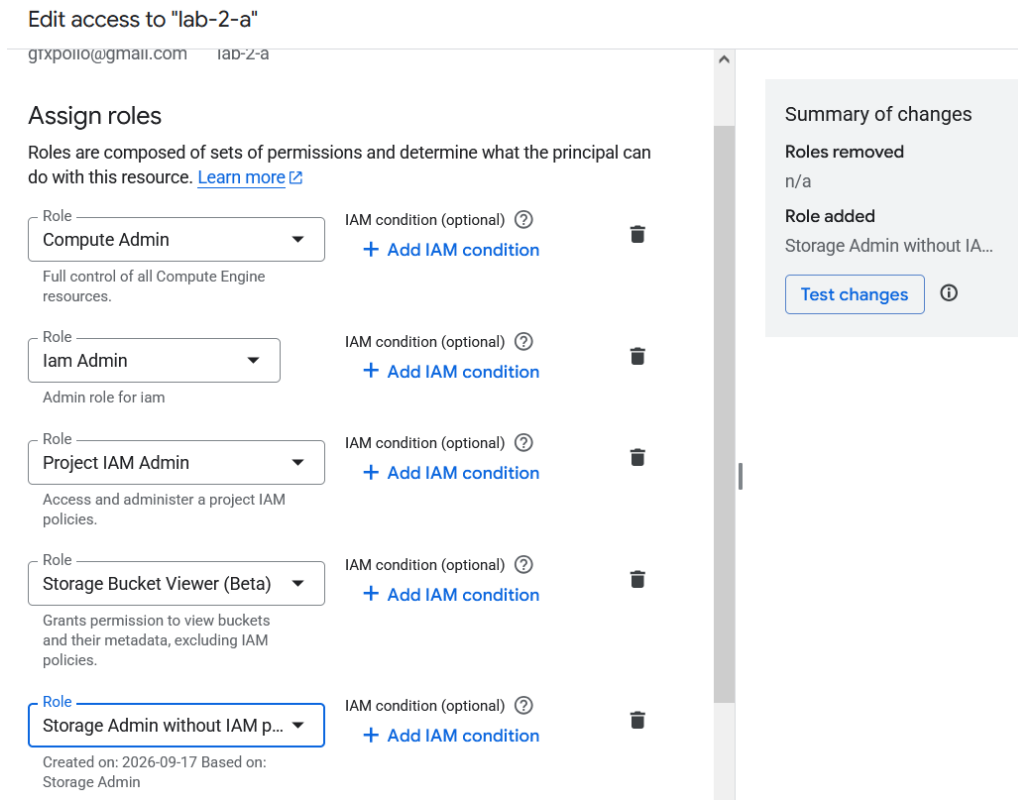


Figure 20: Storage Editor removed and the custom role added

Verification: User-B can still create buckets and objects, but the permissions tab of the bucket no longer lets him grant access to anyone.

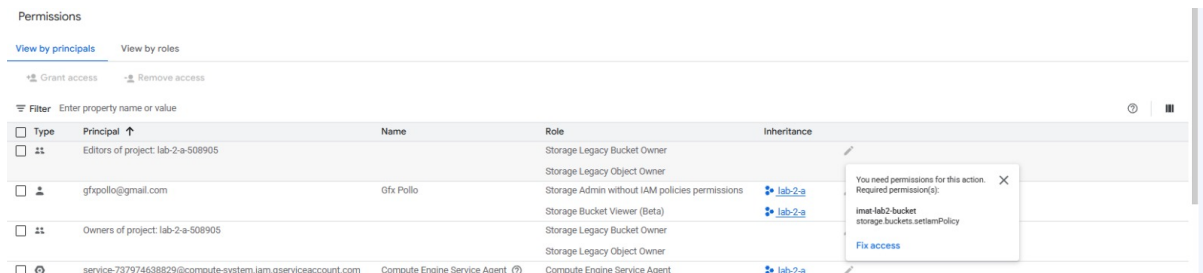


Figure 21: User-B cannot manage permissions on the bucket with the custom role

4 Task 3 – Service accounts

User-A created the service account and granted it Storage Bucket Viewer on the project, so that anything running as that identity can list the buckets.

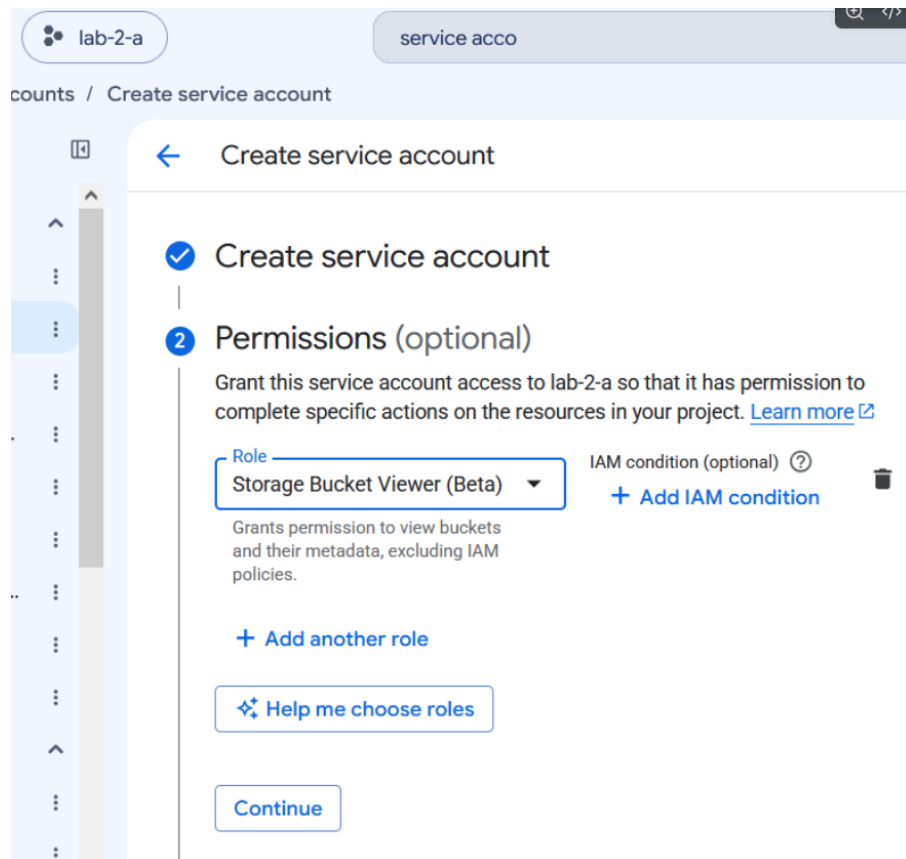


Figure 22: Adding Storage Bucket Viewer Role to the Service Account

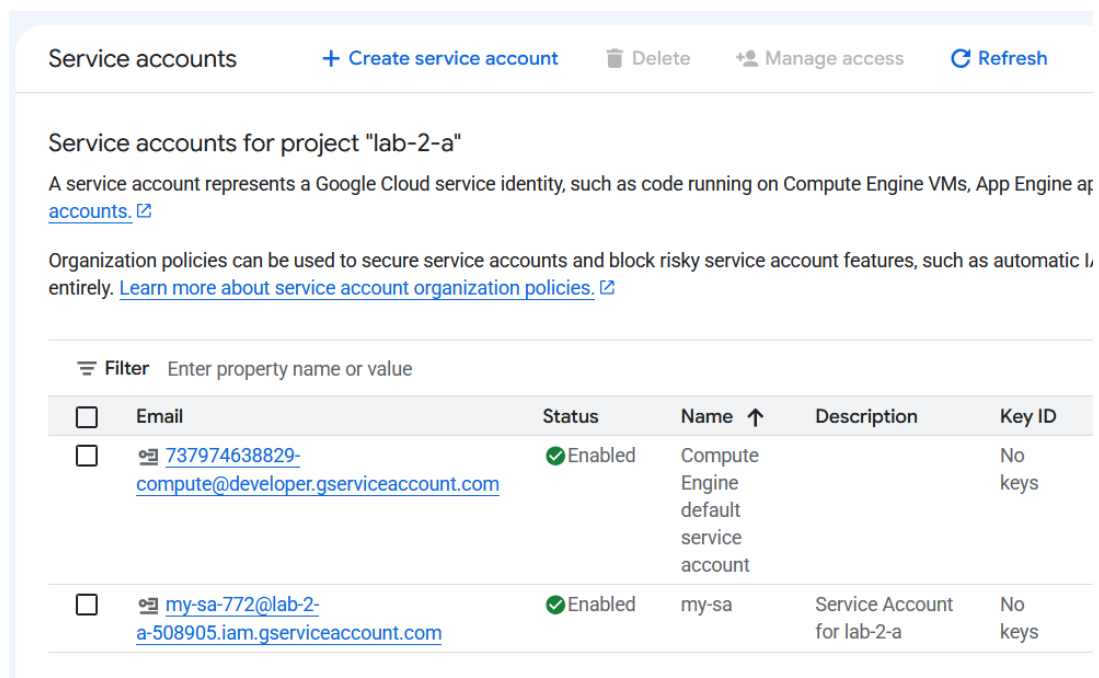


Figure 23: Service account my-sa created in lab-2-a

Then he created vm3 (e2-medium, europe-west1-c, Debian default, tag deep, network ml-network) and attached the service account to it, leaving the default access scopes.

[Create VM from...](#)

Security

Identity and API access [?]

Service accounts [?]

Service account
 my-sa

To access instances with this service account you need to add the Service Account User role: roles/iam.serviceAccountUser. [Learn more](#)

Access scopes [?]

☒ Allow default access

☐ Allow full access to all Cloud APIs

☐ Set access for each API

(a) my-sa attached to vm3

VM instances [Create instance](#) [Import VM](#) [Refresh](#)

[Instances](#) [Observability](#) [Instance schedules](#)

VM instances

[Filter](#) Enter property name or value

☐	Status	Name [↑]	Zone	Recommendations	In use by
☐	✓	vm2	europa-west1-c		
☐	✓	vm3	europa-west1-c		

(b) vm2 and vm3 running

Figure 24: Creation of vm3 with the service account

Q: User-B connects to vm3 and tries to list the buckets in the project. What is the result? Why?

He cannot even connect, so he cannot list anything. The IAP connection fails with *Code: 4033, Reason: not authorized*, because at that moment User-B has no permission to use the Identity-Aware Proxy tunnel (`iap.tunnelInstances.accessViaIAP`); the same lack of `iam.serviceAccountUser` seen in Task 1 also applies, since vm3 runs as my-sa.

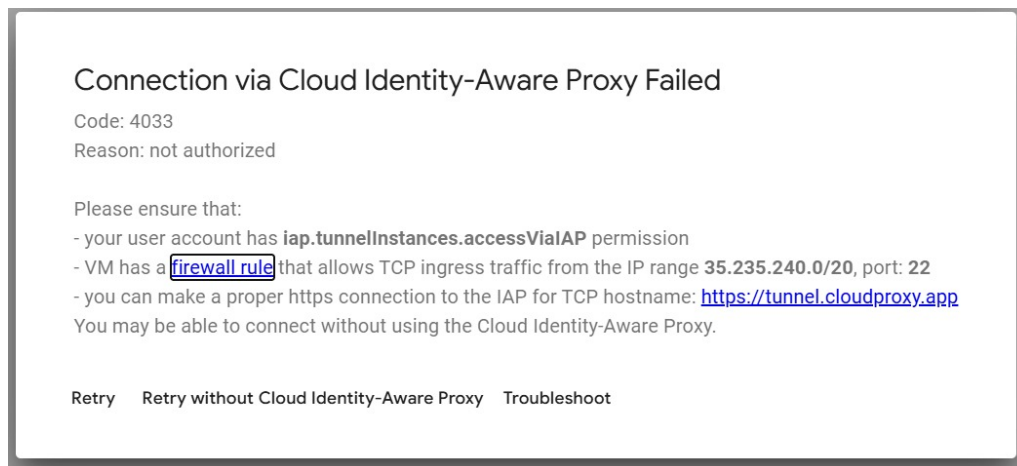


Figure 25: IAP connection to vm3 rejected

Q: What is the role that had to be added to User-B?

IAP-secured Tunnel User (`roles/iap.tunnelResourceAccessor`), which allows reaching the instances through the Identity-Aware Proxy, the only path available because the firewall only accepts SSH from the IAP range.

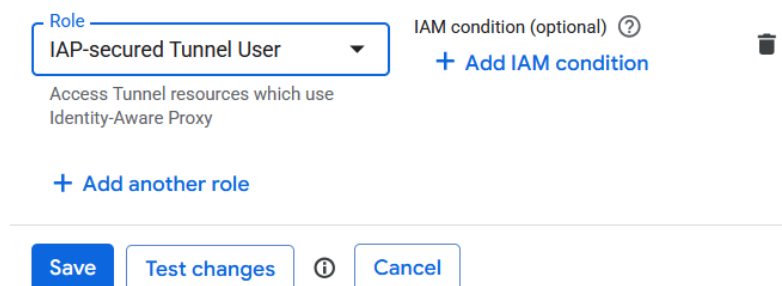


Figure 26: User-A adds IAP-secured Tunnel User to User-B

After the grant, User-B connected to `vm3` and listed the buckets successfully. The command below succeeds because of the identity of the *VM*, not of the user. The CLI inside the instance authenticates as `my-sa`, which holds Storage Bucket Viewer.

Listing 4.1: Listing the buckets from inside vm3

```
gcloud storage buckets list
```

```

gfxpollo@vm3:~$ gcloud storage buckets list
---
cmek_enforcement:
  restrictionMode: NotRestricted
creation_time: 2026-09-17T06:46:55+0000
csek_enforcement:
  effectiveTime: '2026-09-17T06:48:54.730000+00:00'
  restrictionMode: FullyRestricted
default_storage_class: STANDARD
generation: 1789627614767623765
gmek_enforcement:
  restrictionMode: NotRestricted
location: US-EAST1
location_type: region
metageneration: 1
name: imat-lab2-bucket
public_access_prevention: enforced
soft_delete_policy:
  effectiveTime: '2026-09-17T06:46:55.196000+00:00'
  retentionDurationSeconds: '604800'
storage_url: gs://imat-lab2-bucket/
uniform_bucket_level_access: true
update_time: 2026-09-17T06:46:55+0000

```

Figure 27: imat-lab2-bucket listed from vm3 using the service account

5 Task 4 – Roles at resource level

User-A removed every role granted to User-B at project level, leaving only himself and the service account in the policy.

Permissions for project "lab-2-a"

These permissions affect this project and all of its resources. [Learn more](#)

☐ Include Google-provided role grants

View by principals View by roles

Grant access Remove access

Filter Enter property name or value

Type	Principal ↑	Name	Role	Security insights ⓘ	Inheritance
☐		andres.cloud.imat@gmail.com	Andrés Gil Vicente	Advanced security insight	andres-cloud-imat-org 
			Organization Administrator		
			Owner		
			Project Mover	Advanced security insight	andres-cloud-imat-org
			Service Usage Admin	Advanced security insight	andres-cloud-imat-org
☐		my-sa-772@lab-2-a-508905.iam.gserviceaccount.com	my-sa	Storage Bucket Viewer (Beta)	Advanced security insight 

Figure 28: IAM policy after removing all the roles of User-B

To allow the virtual machines to appear in the Compute Engine console, a project-level permission had to be assigned first. A cleaner way to achieve this visibility is by granting a narrow read-only role, such as `roles/compute.viewer`, at the project level.

Once this baseline visibility was established, Compute Instance Admin (v1) (`roles/compute.instanceAdmin.v1`) was granted to User-B on the resource `vm3` only, not on the entire project.

Grant access to "vm3"

Grant principals access to this resource and add roles to specify what actions the principals can take. Optionally, add conditions to grant access to principals only when a specific criteria is met. [Learn more about IAM conditions](#)

Resource

 vm3

 Make sure all fields are correct to continue

Add principals

Principals are users, groups, domains, or service accounts. [Learn more about principals in IAM](#)

New principals *
  

Assign roles

Roles are composed of sets of permissions and determine what the principal can do with this resource. [Learn more](#)

Role *
Compute Instance Admin (v1) 
Full control of Compute Engine instances, instance groups, disks, snapshots, and images. Read access to all Compute Engine networking resources.

IAM condition (optional) 
[+ Add IAM condition](#) 

Save

Cancel

Figure 29: Compute Instance Admin (v1) granted at `vm3` level

Compute Instance Admin (v1) (roles/compute.instanceAdmin.v1)

Full control of Compute Engine instances, instance groups, disks, snapshots, and images. Read access to all Compute Engine networking resources.

If you grant a user this role only at an instance level, then that user cannot create new instances.

Figure 30: Documentation of roles/compute.instanceAdmin.v1

Q: User-B tries to change the configuration of vm3 and tries to create new virtual machines. Is User-B able to do these tasks? Why?

He can edit vm3, but he cannot create new instances. The role gives full control over the instance it is granted on, which covers editing vm3. Creating a machine is a different matter for two reasons. First, the permission to create instances lives at project level, and Google documents that granting this role only at instance level does not allow creating new instances. Second, the attempt failed with an explicit error: the new VM would run as the default Compute Engine service account and User-B has no `iam.serviceAccountUser` on it.

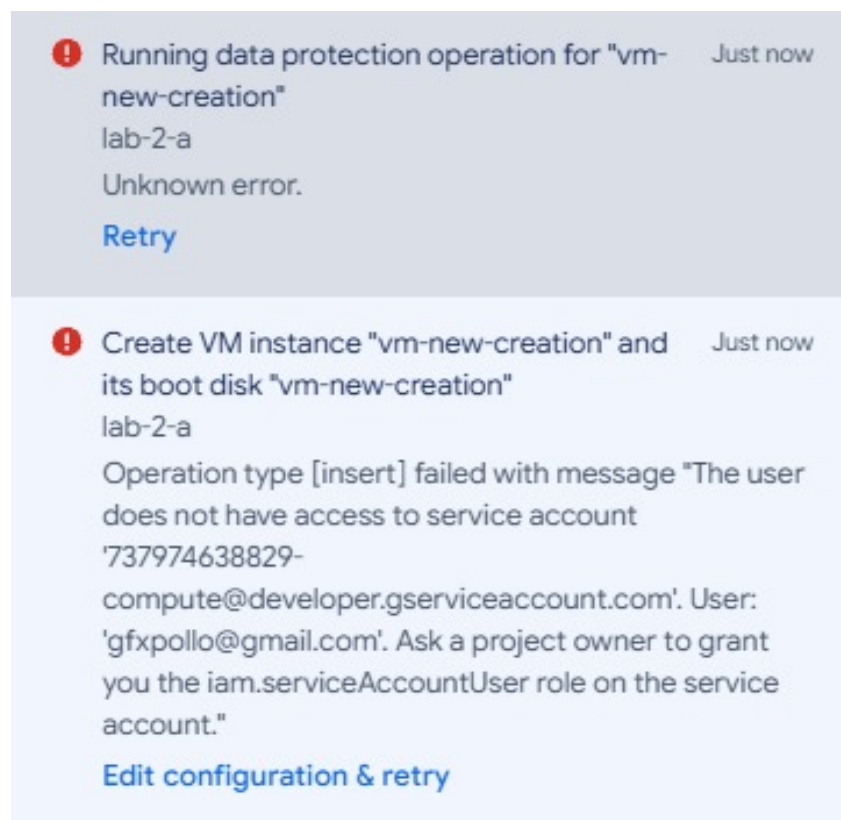


Figure 31: Creating a new VM fails: no access to the default service account

6 Task 5 – IAM best practices (Activity)

Scenario: IT Admins build and maintain a set of curated disk images; SW Developers create Compute Engine instances from those images. The design below follows the least privilege principle.

1. Projects. Two projects instead of one, so that the boundary between infrastructure and development is also an IAM boundary:

- **shared-images-project:** only for building, storing and maintaining the curated images.
- **sw-dev-project:** the working environment where developers deploy and run their instances.

2. Identities. Two Google groups, one for IT and other for SW. And the roles will be granted to the groups, never to individual accounts.

3. Roles.

Group	Project	Role and resulting access
IT Admins	shared-images-project	roles/compute.imageAdmin: create, update, delete and maintain the disk images
IT Admins	sw-dev-project	none: no access to the developer environment
SW Developers	shared-images-project	roles/compute.imageUser: list and use the curated images to launch VMs, but not modify, delete or create images
SW Developers	sw-dev-project	roles/compute.instanceAdmin.v1: create, start, stop and configure their own instances

Table 2: Groups, projects and predefined roles of the proposed design

Compute Image User (roles/compute.imageUser) Permission to list and read images without having other permissions on the image. Granting this role at the project level gives users the ability to list all images in the project and create resources, such as instances and persistent disks, based on images in the project.	compute.images.get compute.images.getFromFamily compute.images.list compute.images.useReadOnly resourceManager.projects.get resourceManager.projects.list serviceUsage.consumerPolicy.analyze serviceUsage.consumerPolicy.get serviceUsage.effectivePolicy.get > serviceUsage.groups.* serviceUsage.quotas.get serviceUsage.services.get serviceUsage.services.list serviceUsage.values.test
Lowest-level resources where you can grant this role:	
• Image	

Figure 32: Permissions contained in roles/compute.imageUser

7 Task 6 – Cost estimation

All figures come from the Google Cloud Pricing Calculator and are monthly costs in USD, which is how the calculator gives them; the yearly cost is simply twelve times that. The baseline is the setup built in this lab: two **e2-medium** instances in **europa-west1** running the whole month, 10 GiB balanced boot disk each, one VPC and one Cloud Storage bucket.

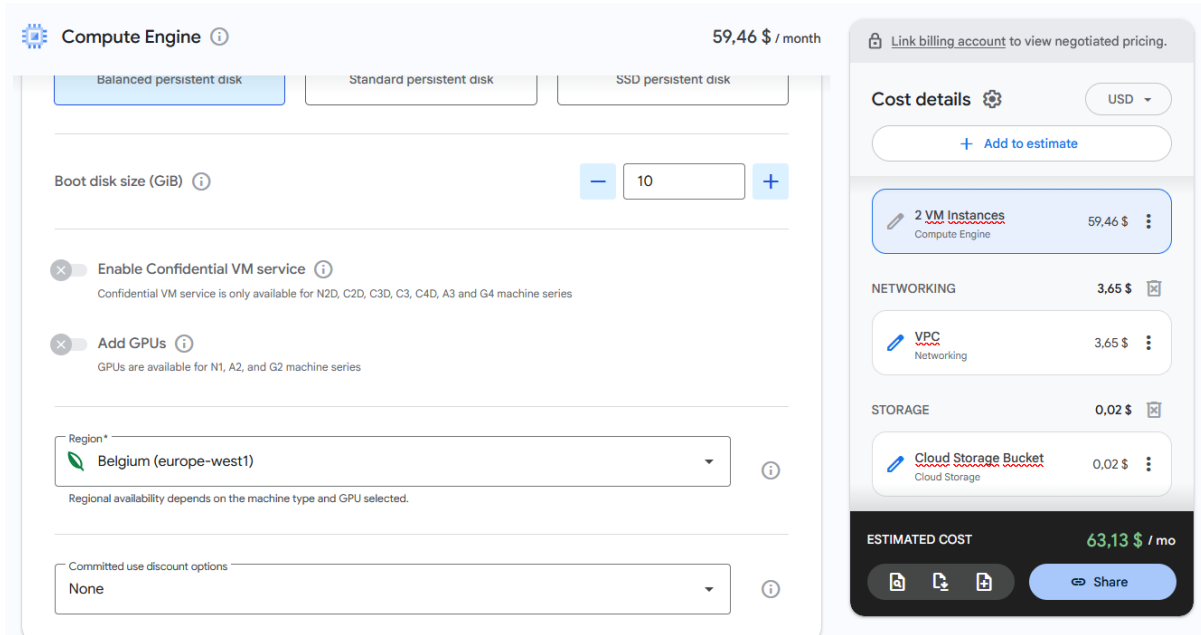


Figure 33: Baseline estimate: $2 \times$ e2-medium, 10 GiB disks, europa-west1

Component	USD / month
2 VM instances (Compute Engine)	59.46
VPC (networking)	3.65
Cloud Storage bucket	0.02
Total	63.13 757.56 USD / year

Table 3: Baseline monthly and yearly cost

7.1 Effect of the configuration

Changing one parameter at a time gives the following totals (networking and storage stay at 3.67 USD/month in every case, so the variation comes entirely from Compute Engine).

We have omitted the screenshots because we think they do not add much information, thanks to the following table:

Change	Configuration	USD / month	USD / year
<i>Baseline</i>			
–	2 × e2-medium, 10 GiB, europe-west1	63.13	757.56
<i>Machine type</i>			
Smaller	e2-small	36.23	434.76
Larger	e2-standard-2	116.94	1 403.28
<i>Boot disk size (measured in us-central1)</i>			
Smaller disk	1 GiB	56.44	677.28
Larger disk	100 GiB	76.24	914.88
<i>Custom machine, vCPUs</i>			
2 vCPU / 4 GiB	custom E2	102.60	1 231.20
6 vCPU / 4 GiB	custom E2	249.73	2 996.76
<i>Region (same machine and disk configuration)</i>			
Hong Kong	asia-east2	77.97	935.64
Madrid	europe-southwest1	67.45	809.40
São Paulo	southamerica-east1	87.97	1 055.64

Table 4: Cost of the setup when changing one parameter at a time

What the numbers show: the machine type dominates, since vCPUs and memory are what is actually billed per second of uptime. Going from **e2-medium** to **e2-small** cuts the bill almost in half, while six vCPUs multiply it by four. Disk size matters much less in absolute terms, around 0.10 USD per GiB and month for a balanced disk, and the region changes the same configuration by roughly $\pm 20\%$, with Madrid close to Belgium and São Paulo clearly more expensive.

Just to take it into account, the two disk-size estimates were taken with the region field set to **us-central1**, so they are comparable to each other but not directly to the baseline. And the vCPU estimates use a custom E2 machine with 4 GiB of memory, not an **e2-medium**.

7.2 Ideas for reducing costs

Q: The workload only needs to run for 2 hours every day.

Start the instances only when the workload runs and stop them afterwards. A stopped VM does not incur VM usage charges, although attached resources such as persistent disks and reserved IP addresses keep costing. Usage would drop from about 730 hours per month and VM to roughly 60, that is, from 1 460 to 120 compute hours for our two machines, more than a tenfold reduction of the compute part of the bill. The start and stop can be automated with instance schedules, so nobody has to remember to turn them off.

Q: The workload needs to run almost non-stop during 3 years.

A 3-year Committed Use Discount fits because the demand is predictable and continuous. Google Cloud charges less in exchange for committing to a minimum level of usage or spend for one or three years. For E2 machines the 3-year flexible CUD is around a 46% discount on eligible vCPU and memory spend. The drawback is that the commitment is paid whether the resources are used or not, so it should only cover the baseline load.

Q: The workload is interruptible and fault tolerant.

Spot VMs. They use the same machine types with discounts of up to 91% over on-demand prices, but Google Cloud can preempt them at any time, so the application has to tolerate interruptions, retry failed tasks and checkpoint its progress periodically. They fit batch processing, simulations and rendering and can be combined with the previous ideas.

8 Conclusions

The lab showed how access in Google Cloud is decided by the combination of who asks, which role they hold and at which level that role was granted. Predefined roles solve most cases if the smallest one is chosen, a custom role is the way to remove a specific permission such as `setIamPolicy` from an otherwise suitable role, and service accounts move the identity from the person to the resource, which is why connecting to a VM needs Service Account User on top of Compute Admin.

Granting a role on a single instance instead of the project made the difference between editing a machine and being able to create new ones very visible.

Finally, the pricing calculator showed that the machine type dominates the cost of a setup like ours, and that scheduling, committed use discounts and Spot VMs are the ways to reduce this cost depending on how the workload behaves.